# Knowledge store: answers and PR #6 review

SkillForge · AI to Data Engineering (DE-01 / DE-05) · 26 October 2026 · From: Ziad Abdullah (AI Lead) · Replaces the earlier embeddings handoff where they differ

> **What changed since the first handoff:** `domain text` is now `domains text[]`, AI reads only from a read-only view, and the view name is configurable (e.g. `knowledge.knowledge_chunks`). Everything else (model, 384 dims, cosine, no prefix on passages) is unchanged.

## 1. Answers to your questions

| Question | Answer |
|---|---|
| Who writes the tables? | DE. Your DE-05 pipeline writes into your own tables (provenance, dedup, cache) and calls our `extract_page()` and `embed()` (or the full `ingest_pages()`). AI only reads. |
| One domain per chunk, or more? | **More than one.** Column `domains text[]` (at least one, no repeats). A chunk appears **once** in the view, not once per domain. |
| Separate schema `knowledge`? Migrations? | Yes. Migrations are DE's, since the tables are yours. Our code reads the name from `SKILLFORGE_PGVECTOR_TABLE`, e.g. `knowledge.knowledge_chunks`. |
| Read tables directly or via an API? | Directly from the view, as a **read-only DB user** (your `skillforge_ai_reader` role is exactly right). Our pgvector store sets every transaction READ ONLY and refuses writes. |
| Retention (raw 30 d, old versions 90 d, crawl logs 12 mo) | Fine. One request: if possible, keep the **clean extracted text** longer than the raw HTML (or until the next crawl), so we can re-chunk or re-embed without re-crawling if the chunk size or model changes. |
| Full model revision for `embedding_version`? | See section 2. |
| `retrieval.md` / `extraction.md` not on any branch? | Sent with this PDF as files, together with `knowledge-chunks.sql`. |

## 2. Pinned embedding model (for embedding_version)

| Field | Value |
|---|---|
| `embedding_model` | `local/BAAI/bge-small-en-v1.5` (384 dimensions) |
| repo | `Qdrant/bge-small-en-v1.5-onnx-Q` |
| revision | `aa8f8b060edb00e03bfdd08813a2949946c8ba55` |
| file | `model_optimized.onnx` |
| sha256 | `51f1bd0addd6e859e42c2c8021a5e5461385bb676a649f4b269aa445449f2431` |

Unchanged rules: vectors are unit length; cosine distance (HNSW `vector_cosine_ops`); passages are embedded with **no prefix**; only search queries get `Represent this sentence for searching`

`relevant passages: .`

## 3. Review of PR #6 (DE-01: Knowledge store schema)

We read `db/schema-draft.sql` and `db/roles-local.sql`. Most of it already matches:

- Column names and types, including `domains text[]` and `char(64)`.
- `embedding_model` checked against our exact string, `vector(384)`, unit vectors, HNSW cosine.
- The `knowledge` schema and the read-only `skillforge_ai_reader` role that can only SELECT the view.

**Three changes we need:**

| # | In PR #6 | Problem | Please change to |
|---|---|---|---|
| 1 | `chunk_id = c.id::text` (a new UUID) | Contract A4 promises the **same id for the same text on the same page**. A new UUID on each re-crawl breaks citations. It passes our format check, so nothing would fail visibly. | Store and expose AI-10's `chunk.chunk_id` (from `extract_page()`). |
| 2 | `title = COALESCE(d.title, '')` | Our model rejects an **empty title**, so a single empty title makes the whole search call fail. | Use the page title from AI-10's `extract_page()` (never empty). |
| 3 | `domains` built by a subquery in a 4-table view with a NOT EXISTS filter | Postgres may then **not use the HNSW index**, so vector search becomes a full scan. | Keep `domains` on the same table as `embedding`; keep the view simple; confirm with EXPLAIN that the HNSW index is used. |

## 4. How the AI code searches

- A row matches query domain D if any of its `domains` equals D or one is a path prefix of the other (`technology` matches `technology/ai`).
- Equal and parent domains use `domains && ARRAY[D and its parents]`, which a **GIN index on domains** can serve. Child domains use `unnest(domains) LIKE 'D/%'`.
- Results are ordered by `embedding <=> query`; score = `1 - distance`.

## 5. Files sent with this PDF

- `knowledge-chunks.sql`: the final DDL generated from our code: base-table example with `domains text[]`, HNSW and GIN indexes, an example `CREATE VIEW knowledge.knowledge_chunks`, the read-only grants, and our search query.
- `retrieval.md`: the full view contract for DE.
- `extraction.md`: includes a new **"For DE-05"** section, from a fetched page to rows in the view.

> Changing the model, the dimension, the `chunk_id` rule or the view columns is a contract change that both AI and DE approve, because it can require rebuilding the index.